H2: Fossil Steam
Methane
Reformation
Methanation and
liquefaction
Grid mix electricity LNG_bCO2_fH2_M_gm
RESOLUTION MEPC.376(80) (adopted on 7 July 2023)
GUIDELINES ON LIFE CYCLE GHG INTENSITY OF MARINE FUELS (LCA GUIDELINES) MEPC 80/17/Add.1
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Order Group Fuel type
Feedstock structure Conversion/Production process Fuel Pathway Code
Feedstock
Type
Nature/Carbon
Source Process Type Energy used in the process
42 LNG Liquefied
Natural Gas
(Methane)
CO2 + H2 CO2: Biogenic
Point Source
Carbon Capture

H2: from
Renewable
electricity
Methanation and
liquefaction
Grid mix electricity LNG_bCO2_rH2_M_gm
43 LNG Liquefied
Natural Gas
(Methane)
CO2 + H2 CO2: Biogenic
Point Source
Carbon Capture

H2: Industrial by-
product hydrogen
Methanation and
liquefaction
Grid mix electricity LNG_bCO2_ibpH2_M_gm
44 CNG Compressed

---CHUNK---

SOURCE: MEPC.376(80).pdf
43 LNG Liquefied
Natural Gas
(Methane)
CO2 + H2 CO2: Biogenic
Point Source

Carbon Capture

H2: Industrial by-
product hydrogen
Methanation and
liquefaction
Grid mix electricity LNG_bCO2_ibpH2_M_gm
44 CNG Compressed
Natural Gas
(Methane)
Natural Gas Fossil Standard refinery
process and
compression
Grid mix electricity CNG_f_SR_gm
45 CNG Compressed
Natural Gas
(Methane)
Mixed 1st,
2nd and 3rd
Gen.
feedstock
Biogenic Thermochemical
gasification
followed by
methanation and
compression
Grid mix electricity CNG_b_G_M_gm
46 CNG Compressed
Natural Gas
(Methane)
Mixed 1st,
2nd and 3rd
Gen.
feedstock
Biogenic Bio-derived LNG
via Anaerobic
Digestion and
separation and
compression
Grid mix electricity CNG_b_AD_gm
47 CNG Compressed
Natural Gas
(Methane)
Mixed 1st,
2nd and 3rd
Gen.
feedstock
Biogenic Bio-derived LNG
via Anaerobic
Digestion,
separation with
Grid mix electricity CNG_b_AD_CCS_gm

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Order Group Fuel type

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SOURCE: MEPC.376(80).pdf
RESOLUTION MEPC.376(80) (adopted on 7 July 2023)
GUIDELINES ON LIFE CYCLE GHG INTENSITY OF MARINE FUELS (LCA GUIDELINES) MEPC 80/17/Add.1
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Order Group Fuel type
Feedstock structure Conversion/Production process Fuel Pathway Code
Feedstock
Type
Nature/Carbon
Source Process Type Energy used in the process
Point Source
Carbon Capture
(PSCC) and long-
term storage and
compression
48 CNG Compressed
Natural Gas
(Methane)
CO2 + H2 CO2: Fossil Point
Source Carbon
Capture

H2: Fossil Steam
Methane
Reformation
Methanation and
compression
Grid mix electricity CNG_fCO2_fH2_M_gm
49 CNG Compressed
Natural Gas
(Methane)
CO2 + H2 CO2: Fossil Point
Source Carbon
Capture

H2: from
Renewable
electricity
Methanation and
compression
Grid mix electricity CNG_fCO2_rH2_M_gm
50 CNG Compressed
Natural Gas

(Methane)
CO2 + H2 CO2: Fossil Point
Source Carbon
Capture

H2: Industrial by-
product hydrogen
Methanation and
compression
Grid mix electricity CNG_fCO2_ibpH2_M_gm
51 CNG Compressed
Natural Gas
(Methane)
CO2 + H2 CO2: Direct Air
Capture

H2: Fossil Steam
Methanation and
compression

---CHUNK---

SOURCE: MEPC.376(80).pdf